
Final Report: Land-Use Classification

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Google Colab Link

Introduction & Illustration

This project classifies RGB satellite image patches from the EuroSAT dataset into 10 land-use categories. Accurate land-use classification is useful for applications such as urban planning, environmental monitoring, and large-scale geographic analysis, but manual labelling of satellite imagery is slow and difficult to scale. Deep learning is a suitable approach because CNNs can learn textures, patterns and spatial features directly from image data, which can allow the model to map an input satellite patch to an output class label accurately and efficiently, as shown in Figure 1.

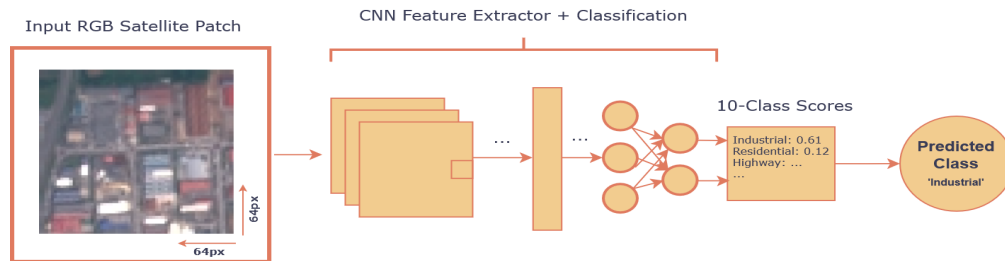


Figure 1: Overview of the land-use classification pipeline

Background & Related Work

Helber et al. introduced EuroSAT, a standard benchmark for land-use classification containing 27,000 labelled Sentinel-2 image patches across 10 classes [4]. Marzougui et al. and Naushad et al. found that CNN-based models, especially when tuned from pre-trained networks, achieve strong performance on EuroSAT but still show the greatest confusion between visually similar classes such as crop and vegetation categories [6] [7]. Jagannathan et al. showed that robustness to seasonal variation can be improved through augmentation strategies in Sentinel-2 segmentation workflows [5]. Fayaz et al. further showed that remote-sensing classification remains challenging in scenes with scale, illumination, and structural variation, and highlighted confusion matrices and saliency-style visualizations as useful tools for interpreting model behaviour [3]. Finally, operational products such as Esri's Living Atlas land-cover maps and Sentinel-2 Land Cover Explorer show the practical value of scalable land-use classification for planning, monitoring, and geographic analysis [2] [1].

Data Processing

I used the EuroSAT RGB dataset and created a stratified 70/15/15 split for training, validation, and testing. For the primary CNN only, I applied label-preserving training-time augmentation using horizontal and vertical flips and 90° and 180° rotations. This increased the training set from 18,900 to 94,500 images, while the validation and test sets were left unchanged to ensure fair evaluation. Figure 2 shows representative class examples and the augmentation types used during CNN training.

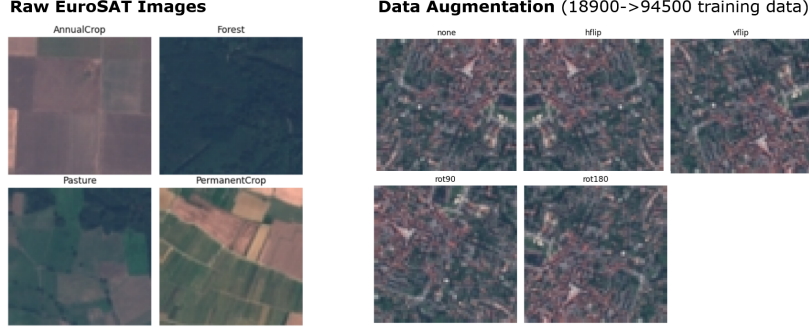


Figure 2: Example of raw EuroSAT data and training-time augmentation

Finally, Table 1 summarizes the final per-class image counts in each split. Table 1 should be read by class-count group rather than by individual class: five classes contain 3,000 images each, four contain 2,500, and Pasture contains 2,000, with the stratified 70/15/15 split and $5\times$ augmentation applied only to the training set.

Table 1: EuroSAT RGB dataset split

Classes	Total	Train	Val	Test	Aug. Train
AnnualCrop, Forest, Herb.Veg., Residential, SeaLake	3,000	2,100	450	450	10,500
Highway, Industrial, PermanentCrop, River	2,500	1,750	375	375	8,750
Pasture	2,000	1,400	300	300	7,000
Total	27,000	18,900	4,050	4,050	94,500

Baseline Model

I used a grayscale histogram of oriented gradients (HOG) feature set followed by a linear support vector machine (SVM) as a baseline model. Each 64×64 RGB image was converted to grayscale, represented using HOG descriptors (9 orientations, 8×8 pixels per cell, 2×2 cells per block), and then classified with a linear SVM. Quantitatively, the baseline achieved 0.5168 validation accuracy, 0.4953 validation macro-F1, 0.5072 test accuracy, and 0.4875 test macro-F1 (Table 2). Further, Figure 3 shows that the model correctly classifies clearer industrial and water scenes, but frequently confuses similar crop and vegetation classes.

Metric	Value
Validation Accuracy	0.5168
Validation Macro-F1	0.4953
Test Accuracy	0.5072
Test Macro-F1	0.4875
Best Class F1 (SeaLake)	0.9137
Worst Class F1 (Pasture)	0.2116

Table 2: HOG + Linear SVM results.

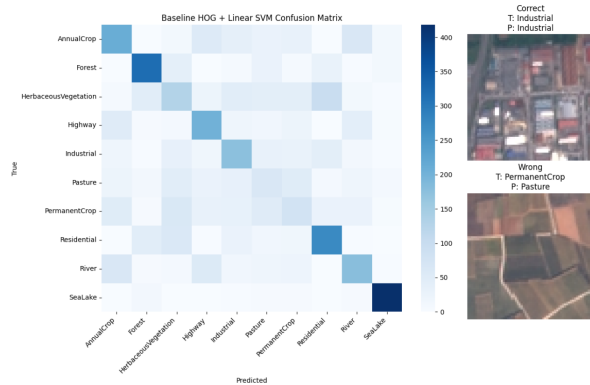


Figure 3: Baseline model confusion matrix.

Primary Model Architecture

The final model is a custom CNN with input size $3 \times 64 \times 64$. It contains three convolutional blocks, each consisting of two 3×3 convolutions followed by batch normalization and ReLU, with channel sizes $32 \rightarrow 32$, $64 \rightarrow 64$, and $128 \rightarrow 128$, and a 2×2 max-pooling layer after each block. This reduces the spatial resolution from 64×64 to 32×32 , 16×16 , and 8×8 , yielding a $128 \times 8 \times 8$ feature map. The classifier head flattens the features to 8192 dimensions, applies a fully connected layer with 256 hidden units, ReLU, dropout ($p = 0.4$), and a final 10-way linear output layer. The full model contains 6 convolutional layers, 6 batch normalization layers, 3 max pooling layers, and approximately 2.39 million trainable parameters. Training used cross-entropy loss, Adam with learning rate 10^{-3} , and batch size 64. Figure 4 illustrates the final model architecture.

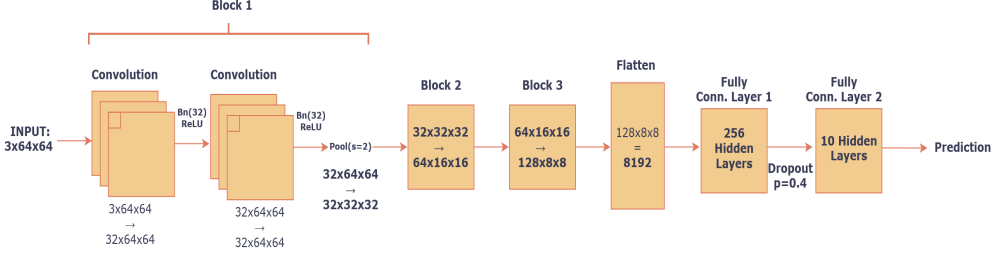


Figure 4: Primary Model Architecture

Evaluation

The final checkpoint was selected with the highest validation accuracy. Final performance was then evaluated once on the test set, which was never used for training or hyperparameter tuning. This provides an estimate of generalization to unseen EuroSAT samples from the same distribution.

Table 3 summarizes the final test set performance of the selected model. Quantitatively, CNN achieved a test accuracy of 0.9694 and a macro-F1 score of 0.9691, while per-class F1 scores ranged from 0.9415 (PermanentCrop) to 0.9856 (SeaLake) as shown in Table 4.

Metric	Value
Test Loss	0.1007
Test Accuracy	0.9694
Test Macro-F1	0.9691
Test Weighted F1	0.9694
Test Macro Precision	0.9703
Test Macro Recall	0.9685
Balanced Accuracy	0.9685
Best Epoch	15

Table 3: Overall test set performance

Class	F1
AnnualCrop	0.9478
Forest	0.9820
HerbaceousVegetation	0.9585
Highway	0.9733
Industrial	0.9853
Pasture	0.9701
PermanentCrop	0.9415
Residential	0.9845
River	0.9628
SeaLake	0.9856

Table 4: Per-class F1 scores

Qualitatively, Figure 5 places the quantitative results in context. The confusion matrix is dominated by diagonal entries, consistent with the model’s test accuracy of 0.9694 and macro-F1 of 0.9691. The selected examples illustrate both correct predictions (first row) and representative failure cases picked from the most common off-diagonal confusion pairs (second row). Correctly classified patches tend to be visually distinctive classes such as Forest, River, and SeaLake, whereas errors occur more often between visually similar agricultural classes such as AnnualCrop and PermanentCrop, which is reflected in the lower per-class F1 for PermanentCrop in Table 4. The examples also show how visually similar these classes can be, and reinforces the need for deep learning in this context.

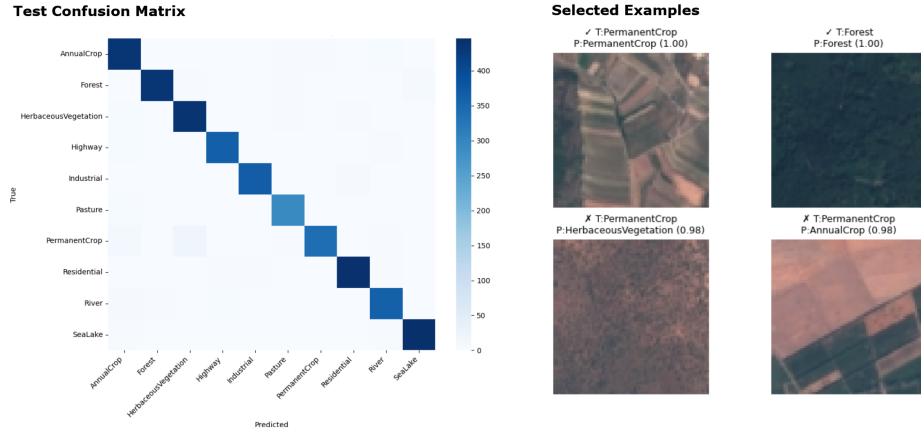


Figure 5: Confusion matrix and select examples of the final model on the test set

Discussion

Overall, the final CNN performed very well on the EuroSAT RGB classification task. The selected checkpoint achieved a high test accuracy, and the close agreement between accuracy, macro-F1, weighted F1, and balanced accuracy suggests strong performance across classes rather than being driven by only a few easy categories. The test confusion matrix is strongly diagonal, further supporting that the model learned useful class-discriminative features.

The per-class F1 scores remained consistently high, but the weaker classes were mostly agriculturally related. As expected, visually distinctive classes such as SeaLake, Forest, Industrial, and Residential achieved the strongest F1 scores, while AnnualCrop, HerbaceousVegetation, and especially PermanentCrop were harder to separate because many patches share similar color, texture, and field geometry. The confusion matrix supports this: the largest off-diagonal errors were concentrated among AnnualCrop, PermanentCrop, and HerbaceousVegetation, with a smaller but interesting confusion between Highway and River! This suggests that the model learned strong class-specific features overall, but errors persist when land-use types with similar textures and layouts overlap.

One interesting result is that the custom CNN trained from scratch performed far better than the HOG + linear SVM baseline. This suggests that learned convolutional features captured spatial texture and layout information that the grayscale HOG representation could not preserve, especially for classes where colour is important. At the same time, the model was still evaluated only on a test split from the same EuroSAT distribution, so the strong performance should be interpreted as strong in-distribution generalization.

Ethical Considerations

A major concern is automation bias. Because the model performs very well overall, users may assume its predictions are reliable in every case and stop questioning edge cases. That is risky for land-use classification because even a small number of errors can matter a lot if the output is used to guide planning, enforcement, or environmental review. Another issue is temporal bias. Satellite images (including the EuroSAT benchmark) are just snapshots, but land use changes over time. A model trained on one distribution of seasons, crop cycles, or development patterns may assign a fixed label to an area that is actually changing. That can create misleading records of dynamic rather than static places. In addition, the dataset contains only 10 single-label classes, which can oversimplify real landscapes where multiple land-use types may appear in the same image patch.

References

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